

QUIDDITY INNOVATIONS

AutoPDF

Local-first PDF toolkit — convert, split, extract, catalog. No cloud.

Components, Architecture & Open-Source Licensing

Technical reference for IT staff

A self-contained, Windows-first PDF toolkit. A tiny PowerShell HTTP server hosts a single-page UI; bundled open-source engines do the document processing; optional intelligence comes from a local Ollama model. Copy a folder, double-click AutoPDF.exe — no installer, no admin, no Python, no telemetry.

PowerShell · localhost:6969 · bundled OSS engines · local Ollama AI

Generated 2026-06-27 · Source: QI Hive Project-Status library

1. What AutoPDF is, for IT

In one line. AutoPDF converts mixed files to PDF in bulk, splits big scanned batches back into individual documents by rules, and extracts metadata from every document into a CSV/XLSX/OnBase-XML index — entirely on the local machine.

Runtime shape. The server is a single ~7,400-line PowerShell script (AutoPDF-Server.ps1) hosting a .NET HttpListener on 127.0.0.1:6969 with ~80 /api/* routes. It targets PowerShell 5.1 syntax so it runs on a vanilla Windows box with nothing installed, and on PS 7+ too. There is no Python runtime and no front-end build step.

Deliberately Python-free & cloud-free. The design choice is 'copy a folder, double-click, done'. All document processing is done by bundled open-source command-line engines; the single AI path is a local Ollama instance — there is no OpenAI/Anthropic/Gemini integration and no telemetry, because confidential documents (legal, HR, registrar, procurement) are a core use case.

Rule-based first, AI second. Splitting and extraction are driven by explicit rules (zones, regex, bookmarks, page ranges). AI only proposes fields and disambiguates — it never silently dictates the schema.

2. Runtime & deployment architecture

Concern	Detail
Server / engine	AutoPDF-Server.ps1 — ~7,400-line PowerShell, .NET HttpListener, hosts UI + ~80 /api/* routes
PowerShell target	5.1 syntax (no ternary / null-coalescing; ASCII hyphens only); 7+ compatible
Bind address / port	127.0.0.1:6969
Launcher	AutoPDF.exe — idempotent C# stub compiled with Windows' built-in csc.exe; pings /api/status, starts server hidden, shows splash, opens browser
Front-end	AutoPDFTool.html single-page shell (7 tabs) + app.js (~5,600 lines vanilla JS) + styles.css (~38 KB)
State	All settings/recipes/templates/library as plain JSON under Application/ (no database)
Bundled engines	Ghostscript, Poppler, Tesseract, Tabula, PDFtk, NAPS2, a bundled JRE — under Application/, invoked as separate processes
AI (optional)	Local Ollama at 127.0.0.1:11434, num_ctx 32768; default model autopdf-mapper:latest
Windows services	QI_AutoPDF (manual placeholder — app runs interactively at logon for Office COM); QI_AutoPDFTunnel (auto-start Cloudflare tunnel)
Public access	autopdf.quiddityinnovations.com via Cloudflare quick tunnel — PIN-gated (/health & /version stay open for probes)
Version stamp	app version 2026.06.15; asset token v=20260624a
Privacy	No telemetry; no cloud document processing; update checks opt-in (VERSION strings only)

3. Component inventory

Every component in the AutoPDF stack. The bundled open-source engines and their licensing obligations are detailed in section 5 — they are the heart of AutoPDF's compliance story.

Layer	Component	What it does / how AutoPDF uses it	License
Server	Windows PowerShell 5.1+	HttpListener on 127.0.0.1:6969 + the entire processing engine; hosts static UI and ~80 /api/* routes.	MIT (PS7) / proprietary (WinPS 5.1)
Server	AutoPDF-Server.ps1	Single-file engine: convert, split, extract, index, OnBase XML, AI proxy, config import/export, scheduler.	Proprietary (QI)
Front-end	Vanilla JavaScript (app.js)	All UI logic (~5,600 lines); no framework, no bundler, no jQuery. State in localStorage.	Proprietary (QI)
Front-end	HTML5 + CSS3	AutoPDFTool.html single-page shell (7 tabs) + hand-written styles.css; zone-mapper.html mapper tool.	Proprietary (QI)
Launcher	C# / .NET (csc.exe)	AutoPDF.exe idempotent launcher (Launcher.cs) compiled with Windows' built-in csc.exe; embedded .ico, splash.	Proprietary (QI)
PDF engine	Ghostscript (gswin64c.exe)	Core PDF rendering / conversion; bundled, called as a separate process.	AGPL-3.0-or-later
PDF engine	Poppler (pdftotext, pdfinfo)	Text + word-bbox extraction for zones, regex, anchors and tagged mapping.	GPL-2.0-or-later
PDF engine	PDFtk (pdftk-java)	PDF page operations / manipulation used by the splitter.	GPL-2.0-only
OCR	Tesseract OCR	Default OCR engine; produces a searchable text layer for scanned / image-only PDFs; multi-language.	Apache-2.0
Tables	Tabula (tabula-java)	True PDF → Excel table extraction.	MIT
Scanning	NAPS2 (NAPS2.Console.exe)	TWAIN / ESCL scanner bridge behind Invoke-ScanCapture; native WIA used otherwise.	GPL-2.0-or-later
Java runtime	OpenJDK / Liberica JRE	Bundled JRE that runs the Java engines (Tabula, pdftk-java).	GPL-2.0-with-classpath-exception
Local AI	Ollama	Optional local LLM host (127.0.0.1:11434) for Smart Mapping chat, AI field proposal, ai_extract fields.	MIT
Local AI	AutoPDF Mapper model	Specialized Ollama variant (AutoPDFMapper.Modelfile) that knows the 3 field kinds + library convention.	Derived from base (e.g. Qwen2.5)
Office	Microsoft Office (COM)	Optional: Word/Excel/PowerPoint and styled Markdown conversion via COM automation.	Commercial (Microsoft)
Index export	Excel COM (XLSX)	Optional XLSX mirror of the CSV index.	Commercial (Microsoft)
Index export	OnBase XML Index DIP	Hyland-schema XML index	N/A (output format)

		output (Write-IndexXml); ConvertTo-XmlName for safe element names.	
Persistence	JSON config files	autopdf-settings.json, recipes/*.json, templates/*.json, regex_library.json, scheduler- jobs.json; no database.	N/A
Networking / ops	Cloudflare Tunnel + NSSM	QI_AutoPDFTunnel (auto- start) exposes PIN-gated public access; tunnel_manager.py + nssm.exe.	Cloudflare ToS / NSSM public domain
Testing	Pester-style PS tests + Playwright	Tier 1/2 PowerShell API/AI tests (~27) and Tier 3 pytest + Playwright UI regression.	Apache-2.0 / MIT
OCR roadmap	RapidOcrNet (ONNX Runtime)	Planned second OCR engine (PP-OCRv5/v6) via .NET ONNX binding; no Python; CPU/CUDA selectable.	Apache-2.0 (planned)

4. The template data model (how extraction works)

Templates are user-defined JSON in `Application/templates/`. Each field is one of three kinds, run in three passes by `Invoke-TemplateExtraction` (`zones` → `regex_match` → `ai_extract`), each priming the next. AI proposes fields but never dictates the schema.

Field kind	What it is	Engine behind it
zone	A fixed rectangle stored as page fractions — pull whatever text sits in that box.	Poppler word boxes
regex_match	A page-text scan with scope (all/first/last/N/A-B) and strategy (first/last/largest/all).	Poppler text + <code>regex_library.json</code>
ai_extract	The local LLM finds the value at run-time when rules can't.	Ollama (autopdf-mapper)
tagged (anchor-relative)	A field that follows its label across documents instead of fixed coordinates; the server re-finds the label on every document and floats the value box beside it.	Find-AnchorRectFrac + Resolve-TaggedZoneRect over Poppler word boxes

A per-field format post-processor (Format-FieldValue) normalizes currency / date / number / phone / upper / lower / trim. The centralized regex library ships ~30 protected builtins (SSN, EIN, phone, dates, currency) and is user-extendable from the Settings tab.

5. Open-source engines pulled from GitHub — the licensing story

The key fact for IT/legal. Ghostscript (AGPL), Poppler (GPL), Pdftk (GPL) and NAPS2 (GPL) are strong-copyleft. AutoPDF stays proprietary because it does NOT link them in-process: every engine lives under `Application/` and is invoked as a separate command-line process (an arm's-length / 'mere aggregation' relationship). The copyleft licenses are not triggered by AutoPDF's own source.

The only obligation is on distribution. Ship the licenses/ folder with the per-engine license files, and do not modify the GPL/AGPL binaries. AutoPDF ships the engines unmodified.

Engine / library	Upstream	License	What it does	How AutoPDF stays compliant
Ghostscript	github.com/ArtifexSoftware/ghostpdl	AGPL-3.0	PDF/PostScript rendering & conversion core.	Separate process (aggregation). Ship license; don't modify binary. Commercial license available from Artifex if ever embedded.
Poppler	gitlab.freedesktop.org/poppler/poppler	GPL-2.0+	PDF text + word-bbox extraction (<code>pdftotext/pdftinfo</code>).	Separate process. Ship license; unmodified binary.
Pdftk (pdftk-java)	gitlab.com/pdftk-java/pdftk	GPL-2.0	PDF page split/merge/manipulation.	Separate process. Ship license; unmodified binary.
NAPS2	github.com/cyanfish/naps2	GPL-2.0+	TWAIN/ESCL scanner capture (console).	Separate process via NAPS2.Console.exe. Ship license; unmodified binary.
Tesseract OCR	github.com/tesseract-ocr/tesseract	Apache-2.0	OCR — searchable text layer for scanned PDFs.	Permissive — keep NOTICE/attribution.
Tabula (tabula-java)	github.com/tabulapdf/tabula-java	MIT	PDF table → Excel extraction.	Permissive — attribution only.

OpenJDK / Liberica JRE	bell-sw.com/ibericajdk	GPL-2.0 + Classpath Exception	Bundled Java runtime for Tabula & pdftk-java.	Classpath Exception — bundling/running a JRE is fine. Ship license.
Ollama	github.com/ollama/ollama	MIT	Optional local LLM host for AI features.	Permissive — attribution only. Separate local service; opt-in.
cloudflared	github.com/cloudflare/cloudflared	Apache-2.0	Cloudflare Tunnel client for PIN-gated public access.	Permissive (client). Cloudflare service under Cloudflare ToS.
NSSM	nssm.cc	Public domain	Runs Ql_AutoPDFTunnel as a Windows service.	Public domain — no obligations.
Playwright / pytest	github.com/microsoft/playwright - github.com/pytest-dev/pytest	Apache-2.0 / MIT	Tier-3 UI regression & test framework (dev only).	Dev/test dependency — not shipped to end users.

6. Feature → component mapping

How the bundled engines and AI tie to what a records team actually does in AutoPDF.

User-facing feature	Components that power it	How
Bulk-convert Office/image/text/Markdown → PDF	Microsoft Office COM, Ghostscript, Pillow-class image handling	/api/run batch-converts an input folder; Office via COM, images/text/Markdown to PDF.
Split one giant scan into many documents	Poppler (word boxes), PDFtk, split runner	9-panel rule-based splitter: zones, patterns, bookmarks, page ranges; renders pages for the visual mapper.
A field that follows its label across docs	Poppler word boxes, Find-AnchorRectFrac / Resolve-TaggedZoneRect	Tagged (anchor-relative) mapping re-finds the label on every document and floats the value box beside it.
OCR a scanned, image-only PDF	Tesseract OCR	Produces a searchable text layer; multi-language select in Settings + Splitter.
Extract metadata into a CSV/XLSX index	Index Creator, Excel COM, regex_library.json	Builds columns+rows from folder tree / filename parts / in-PDF content; FILENAME leads every index.
Hand the index to OnBase	Write-IndexXml (OnBase XML Index DIP)	Emits a Hyland Foundation 24.1 schema XML DIP (one <NewDocument> per row), UTF-8 no-BOM, XML-escaped.
Design a template by chatting with AI	Ollama (autopdf-mapper), /api/ai-chat, /api/template-propose	Page-range-aware chat; AI proposes fully-configured fields + regex-library additions — opt-in, fully local.
Capture straight from a scanner	NAPS2 (TWAIN/ESCL) + native WIA	Invoke-ScanCapture dispatches to NAPS2 or falls back to native WIA.
Pull tables out of a PDF	Tabula (+ bundled JRE)	True PDF → Excel table extraction.
Reach AutoPDF remotely, safely	Cloudflare Tunnel + NSSM, access-PIN gate	Ql_AutoPDFTunnel exposes PIN-gated access; /health & /version stay open for probes only.

7. Licensing & compliance posture (summary for IT)

Copyleft handled by aggregation. The strong-copyleft engines (Ghostscript AGPL; Poppler, PDFtk, NAPS2 GPL) are invoked as separate processes, never linked in-process — so they do not impose their license on AutoPDF's proprietary source. Obligation: distribute the unmodified binaries with their license files.

Permissive remainder. Tesseract (Apache-2.0), Tabula (MIT), Ollama (MIT), cloudflared (Apache-2.0) and the bundled Liberica JRE (GPL + Classpath Exception, which explicitly permits bundling) are all free-to-use; obligations are limited to keeping attribution/NOTICE files.

Privacy by construction. No telemetry; no cloud document processing; loopback bind by default; PIN-gated when tunnelled. The only network calls are opt-in update checks that fetch VERSION strings only.